

Pranav Srinivasan

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EDUCATION

Kent State University

Bachelor of Science in Physics

Kent, OH

Expected May 2027

- **Relevant Coursework:** Thermodynamics, Quantum Mechanics, Control Systems, MATLAB Programming

RESEARCH EXPERIENCE

Undergraduate Researcher

Dr. Thorsten Schmidt Lab, Kent State University

Oct 2023 – Present

Kent, OH

- Investigated DNA origami structures, focusing on support structures to suspend DNA nanodiscs for use in membrane protein imaging and structure characterization.
- Utilized CaDNano to design DNA origami structures. Developed skills in micropipetting, DNA origami synthesis, and analysis using gel electrophoresis and Transmission Electron Microscopy (TEM).

ENGINEERING PROJECTS

HAVOC Flight Computer

Embedded Engineer / Contributor

Aug 2025 – Present

- Designed and routed custom PCB boards for various flight computer subsystems.
- Integrated environmental sensor suites (BMI088, MS5611, BME688, SHT41) for high-altitude telemetry collection.

IREC High-Power Rocketry

Club-Side Payload Lead / Contributor

Aug 2024 – Present

Kent, OH

- Lead payload development and testing of the PCBs and electronics for a 4U, weather station payload driven by a Raspberry Pi and Arduino.
- Developed the software stack to interface with the sensors and control the payload's landing legs mechanism. Analysis of environmental data on-board the Raspberry Pi to calculate atmospheric conditions including wind speed and direction.

Amateur Radio & Remote Telemetry Beacon

Hardware & Software Implementation

Aug 2025 – Present

Kent, OH

- Configured a Raspberry Pi 3A+ running a Python-based telemetry stack for AX.25 packet decoding and APRS transmission.
- Interfaced HackRF and RTL-SDR hardware with SDR++ for real-time digital communications and signal analysis.
- Developed feed-horns for a repurposed WiFi dish to downlink MetOp and NOAA GOES satellite image data.
- Developed Yagi-Uda antennas for uplink and downlink of images and telemetry data.

AstroFlashes, FlashSat-1

Deputy Lead, Communications

Aug 2025 – Present

Kent, OH

- Lead the communications team for the AstroFlashes project, managing the on-board telemetry and APRS transmission.
- Developed F Prime (F') stack for on-board telemetry and APRS transmission.
- Laid the groundwork for ground station development at Andrew Paton Airport
- Helped development of a buildroot operating system for the BeagleBone Black Industrial
- Development of Student Research and Developed (SRAD) PCB carrier boards for the BeagleBone Black Industrial and Raspberry Pi Compute Module 5

TECHNICAL SKILLS

Languages: C, C++, Python, L^AT_EX, Bash

Hardware & Embedded: KiCad, Arduino, Raspberry Pi, BeagleBone, SDR, Oscilloscopes, MMANA-GAL, MATLAB, Transmission Electron Microscopy (TEM)

Software & Tools: Arch Linux (Hyprland), Fedora, Ubuntu, Neovim, F Prime (F'), Git, CaDNAno